Funding Shocks in a Multi-Factor Risk Equilibrium: Direct-Yield Evidence on Japanese–Indonesian Transmission

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Abstract

A popular reading of the 2024–2026 yen carry unwind is that rising Japanese Government Bond (JGB) yields drove emerging-market currencies, the Indonesian rupiah among them, into disorderly weakness. Using ten years of *direct* 10-year yield data for Japan and Indonesia—rather than the ETF proxies common in this literature—we put that reading to a quantitative test inside a five-variable cointegrated system that adds the dollar index and the VIX. Three results reshape the story. First, the rupiah's long-run equilibrium is not bilateral: a stable cointegrating space exists only once the VIX is included (the Johansen trace rank rises from zero to two), so risk appetite is a structural component of the equilibrium, not an external control. Second, a forecast-error variance decomposition shows that Japanese funding shocks explain only about 3% of rupiah variance at a 30-day horizon, against roughly 30% for the dollar and a rising 15% for the VIX—and the 3% figure holds even when JGB is ordered first, the most generous identification available to it. Third, a Monte Carlo built from the estimated VECM dynamics (not a random walk) error-corrects back toward equilibrium rather than overshooting; the realized move of USD/IDR toward 18,000 sits above the model's long-run anchor and is better read as a multi-factor repricing than a single-driver overshoot. Japanese funding shocks contribute to rupiah dynamics, but their effect operates primarily through a broader equilibrium system dominated by global dollar conditions and risk appetite.

Keywords: carry trade, cointegration, variance decomposition, risk appetite, exchange rates, Indonesia

1 Introduction

When the Bank of Japan began letting long-term yields rise, the carry trade that had quietly funded a decade of emerging-market positions started to turn. The mechanics are familiar: a higher JGB yield raises the cost of the yen funding leg, compresses the carry, and prompts an unwind in which borrowed yen is repurchased and the assets bought with it are sold. The rupiah, liquid and widely held, is a natural place for that selling to land. By mid-2026 USD/IDR had pushed toward 18,000, and the obvious narrative wrote itself—Japanese yields went up, the rupiah went down.

We think that narrative is half right in a way that matters. Japanese funding shocks do transmit to Indonesian markets; we confirm it three different ways. But "transmit" and "drive" are not the same claim, and the difference is the point of this paper. A variable can have a statistically significant, economically sensible effect and still account for very little of what actually moves a price. Separating the existence of a channel from its quantitative weight is exactly what a carry-unwind episode demands, because the same months that contained the JGB surge also contained a stronger dollar, a global repricing of risk, portfolio outflows from emerging markets, and Indonesia-specific pressures. Attributing the rupiah's move to any one of these without measuring the others is guesswork dressed as inference.

Two design choices let us do better than the existing applied work on this question. First, we use direct 10-year government bond yields for both Japan and Indonesia over 2016–2026, rather than the equity or bond ETF proxies that earlier studies (including an earlier version of our own) lean on when yield data is hard to assemble in real time. Proxies capture risk sentiment but blur the yield channel they are meant to measure; direct yields let us read coefficients as transmission rather than as a sentiment correlation. Second, we embed the two yields in a five-variable system alongside the dollar index and the VIX, and we let the data, not convention, decide the cointegration rank.

Those choices produce a result we did not expect when we started. The bilateral JGB–rupiah relationship barely cointegrates; the system only settles into a stable long-run equilibrium once risk appetite enters it. And once the full system is estimated, a variance decomposition puts Japanese funding shocks at the bottom of the list of what moves the rupiah, behind the rupiah's own idiosyncratic dynamics, the dollar, and the VIX. The transmission is real. It is also small, and it operates inside a risk-cycle equilibrium that the dollar and risk appetite dominate. That is a more honest and, we think, more useful account of how a funding shock in Tokyo reaches a currency in Jakarta.

2 Literature Review

The econometric frame is the cointegration tradition. Granger (1981) established that integrated series can share stationary linear combinations, the formal basis for treating a long-run currency–yield relationship as an equilibrium rather than a spurious regression, and Engle & Granger (1987) gave that idea its error-correction representation. Johansen (1991) extended estimation to a multivariate maximum-likelihood setting with the trace test we use to fix the cointegration rank. For decomposing where variance actually comes from we follow Sims (1980), whose orthogonalized variance decomposition is the natural tool for asking how much of an exchange rate's movement each shock explains, and we rely on Lütkepohl (2005) for the VECM and forecast-error-variance-decomposition machinery.

On the economics, the carry trade is well charted. Brunnermeier et al. (2009) tie carry returns to currency-crash risk and the fat-tailed losses that arrive during unwinds, while Gabaix & Maggiori (2015) supply an intermediary mechanism in which constrained financial intermediaries make exchange rates sensitive to funding and capital-flow shocks—the channel through which Tokyo reaches Jakarta. The contagion literature supplies the caution we keep throughout: Forbes & Rigobon (2002) distinguish genuine contagion from ordinary interdependence and warn that cross-market correlation rises mechanically with volatility, which is precisely why we lean on variance shares rather than correlations to weigh the JGB channel. The Bayesian step follows Gelman et al. (2013) in how we report posterior credibility.

3 Data and Methodology

3.1 Data

We use daily data from May 2016 through June 2026, 2,650 aligned trading-day observations. The two yields are *direct* 10-year government benchmarks: the JGB 10-year and the Indonesian SBN 10-year, sourced from market data rather than proxied by ETFs. We add three variables that the carry-unwind narrative implicates: USD/IDR, the broad trade-weighted U.S. dollar index, and the VIX. A WTI oil series enters later as a robustness control for Indonesia's terms of trade.

This is a deliberate departure from our earlier proxy-based work. That version used the TOPIX ETF and an Indonesian equity ETF as stand-ins for the two yields because direct series were not accessible in real time; it carried an explicit caveat that replication with direct yields was needed. This paper is that replication. Using real yields, the coefficients below can be read as transmission of the yield itself, not of an equity-market sentiment correlated with it.

3.2 Stationarity

Augmented Dickey–Fuller tests (Table 1) place the JGB yield, USD/IDR, the dollar index, and oil firmly in the $I(1)$ camp. Two series sit awkwardly: the SBN yield rejects the unit root at the 5% level ($p = 0.012$) and the VIX rejects it decisively ($p < 0.001$). We treat both as *near-$I(1)$*—persistent, mean-reverting only slowly, behavior that is typical of a managed bond market and a volatility index—and we verify in Section 5 that the central results survive a stationary first-difference treatment. The key inferences below do not rest on a strict unit-root assumption for these two.

Table 1: Augmented Dickey–Fuller tests on level series.

Series	ADF p-value	Classification
JGB 10Y	1.000	$I(1)$
SBN 10Y	0.012	near-$I(1)$
USD/IDR	0.837	$I(1)$
DXY	0.214	$I(1)$
VIX	0.000	near-$I(1)$

3.3 Lag order and cointegration rank

We select the lag from an unrestricted VAR using all four standard criteria rather than AIC alone. AIC favours six lags while BIC and HQIC favour two; we carry the AIC choice of six into the VECM as $k_{\text{ar diff}}$ and confirm in Section 5 that the conclusions are stable across this range. We then fix the cointegration rank from the Johansen trace test rather than imposing it, estimating the test for two systems: the bilateral-plus-dollar baseline $[\text{JGB}, \text{SBN}, \text{DXY}, \text{USDIDR}]$ and the same system augmented with the VIX.

3.4 VECM, variance decomposition, and simulation

The system is written in error-correction form,

$$\Delta \mathbf{y}_t = \boldsymbol{\alpha}\, \boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{i=1}^{k} \boldsymbol{\Gamma}_i\, \Delta \mathbf{y}_{t-i} + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t, \qquad (1)$$

estimated by maximum likelihood with a restricted constant inside the cointegration space. The loadings $\boldsymbol{\alpha}$ identify which variables adjust toward equilibrium and which are weakly exogenous. To weigh the channels we compute the orthogonalized forecast-error variance decomposition (FEVD) of USD/IDR, which apportions its forecast variance at each horizon among the system's shocks. We report the central case with the variables Cholesky-ordered from most exogenous (JGB) to most endogenous, the ordering most generous to the JGB channel, and treat robustness to ordering in Section 5.

A Monte Carlo experiment then propagates the system forward. Crucially, and unlike a naive random-walk projection, we simulate from the *estimated VECM dynamics*: each path evolves under $\Delta \mathbf{y}_t = \boldsymbol{\alpha}(\boldsymbol{\beta}'\mathbf{y}_{t-1} + c) + \sum_i \boldsymbol{\Gamma}_i \Delta \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t$ with innovations drawn from the residual covariance, so error correction and lag structure are present. We run an unconditional version and a conditional version in which the JGB path is exogenously raised, a stylized BoJ-normalization scenario. Finally, a Bayesian regression on daily changes provides a directional consistency check on the JGB–SBN link, reported following Gelman et al. (2013).

4 Results

4.1 Risk appetite completes the equilibrium

The Johansen results (Table 2) are the first surprise. The baseline four-variable system does not cointegrate at conventional levels: the trace statistic for no cointegration, 46.6, falls below the 5% critical value of 47.9. There is no stable bilateral JGB–rupiah–dollar equilibrium to speak of. Add the VIX and the picture changes completely—the trace statistic jumps to 92.4 and the system supports a cointegration rank of two at the 1% level. Risk appetite is not a control we bolt onto the side of the model; it is the variable that makes a long-run equilibrium exist at all.

Table 2: Johansen trace test: the VIX completes the cointegrating space.

	Baseline (4-var)		+ VIX (5-var)	
H_0	Trace	95% CV	Trace	95% CV
$r \leq 0$	46.6	47.9	92.4	69.8
$r \leq 1$	20.8	29.8	48.4	47.9
$r \leq 2$	8.7	15.5	21.9	29.8
Rank @5%	0		2	

The cointegrating vectors confirm that the VIX belongs inside the relation rather than beside it. Because the variables live on very different scales, we read the vectors through each term's contribution—its coefficient times the variable's mean level. In the first relation the VIX contributes 1.61 against the JGB's 0.43; in the second it contributes 2.59. Risk appetite carries weight in the long-run equilibrium comparable to, and in places larger than, the funding yield itself.

The loading coefficients (Table 3) place the burden of adjustment squarely on the rupiah. USD/IDR carries large, highly significant loadings (12.93 and −3.97); it is the variable that moves to restore equilibrium. The VIX itself error-corrects on the second relation ($\alpha_2 = -0.124$, $p < 0.01$), reinforcing that it is part of the system's dynamics, not an exogenous bystander. The JGB yield adjusts weakly at best.

Table 3: VECM loading coefficients ($\boldsymbol{\alpha}$), five-variable system.

Equation	α_1	p	α_2	p
JGB 10Y	0.0018	0.090	−0.0014	0.005
SBN 10Y	0.0076	0.005	−0.0020	0.110
DXY	0.0150	0.424	0.0050	0.563
USD/IDR	**12.93**	**0.000**	−3.97	**0.002**
VIX	−0.139	0.156	−0.124	0.007

4.2 What actually moves the rupiah

The variance decomposition (Figure 1, Table 4) is the heart of the paper. At a 30-day horizon, Japanese funding shocks explain about 3% of USD/IDR forecast-error variance. The dollar explains roughly ten times that, near 30%, and risk appetite explains close to 15% and rising—the VIX share climbs from 1% at a one-day horizon to 15% at thirty days, exactly the signature of a variable that matters more the longer the horizon, consistent with its place in the long-run equilibrium. The rupiah's own idiosyncratic shocks account for the largest share, around 44%.

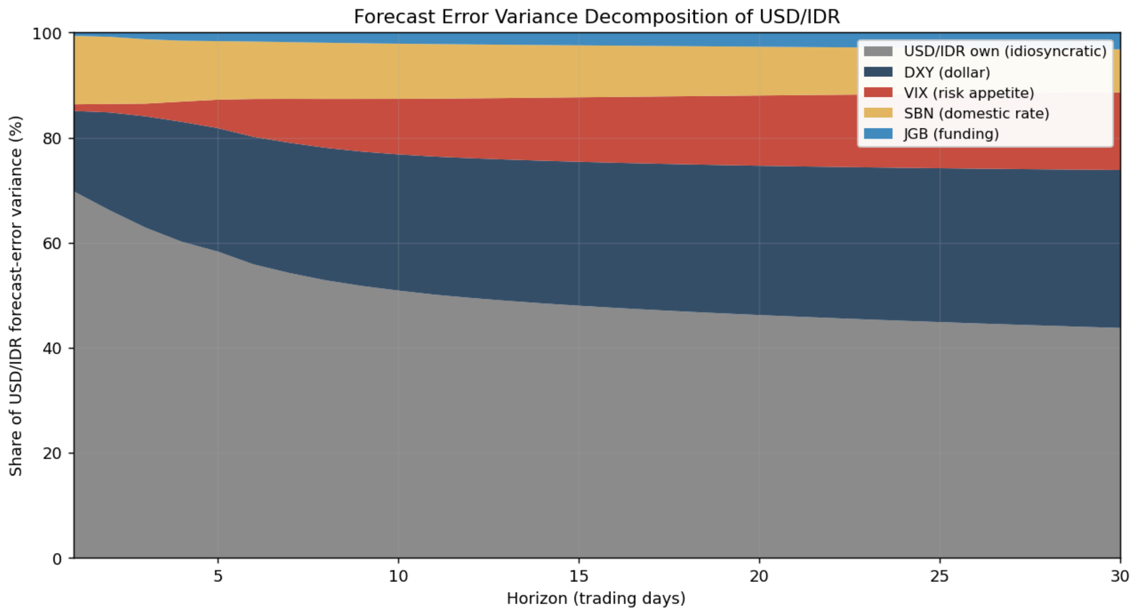

Figure 1: Forecast-error variance decomposition of USD/IDR. Japanese funding shocks (top band) are a thin sliver; the dollar and risk appetite dominate the external contribution, and the VIX share grows with the horizon.

The 3% figure deserves emphasis because it is conservative by construction. We ordered the JGB first in the Cholesky decomposition, which hands it the maximum share of any contemporaneously correlated variance. Even granted that advantage, it explains

Table 4: FEVD of USD/IDR by horizon (% of forecast-error variance).

Horizon	USD/IDR own	DXY	VIX	SBN	JGB
1 day	70	15	1	13	1
10 days	51	26	11	11	2
30 days	44	30	15	8	3

almost nothing. A funding shock in Tokyo is a genuine but minor input to the rupiah; the currency is governed by the dollar and the global risk cycle.

4.3 Impulse response and directional check

The transmission, though small in variance terms, is real and signed as theory predicts. A one-standard-deviation JGB shock pushes USD/IDR up—a weaker rupiah—and the response is persistent, with the lower confidence band staying above zero across the horizon (Figure 2). The response of the SBN yield is similar but its lower band dips toward zero around day six before recovering, consistent with a two-phase reaction: an immediate transmission phase over days zero to five and a slower repricing phase as less active participants rebalance.

A Bayesian regression of daily SBN changes on daily JGB changes corroborates the direction with explicit uncertainty (Figure 3). The posterior mean of the slope is 0.28 with essentially all of its mass above zero; adding the dollar as a control leaves it at 0.26. We frame this narrowly: it is a directional consistency check on the bilateral channel, not a confirmation of the full system, which the VECM captures. The bivariate specification intentionally isolates the JGB–SBN link.

4.4 Monte Carlo: error correction, not overshoot

Simulating from the estimated VECM dynamics gives a result the random-walk projections in this literature miss (Figure 4). Because the rupiah is the system's adjuster, the unconditional simulation *error-corrects*: starting from 17,690, the median path drifts down toward roughly 16,200, the model's long-run anchor, rather than running to 18,000. Raising the JGB along an exogenous path shifts the median higher, to about 16,700—the channel pushes in the right direction—but it does not reach 18,000 within the horizon.

The reading is straightforward and, we think, more defensible than the overshoot story. The model identifies the transmission channel correctly, but the realized 18,000 episode is associated with a shift in equilibrium conditions rather than a temporary overshoot around a fixed equilibrium. The rupiah in mid-2026 sat above the equilibrium implied by a decade of data, which is what one expects when several forces—a normalizing BoJ, a firm dollar, and a global risk repricing—push in the same direction at once. A single

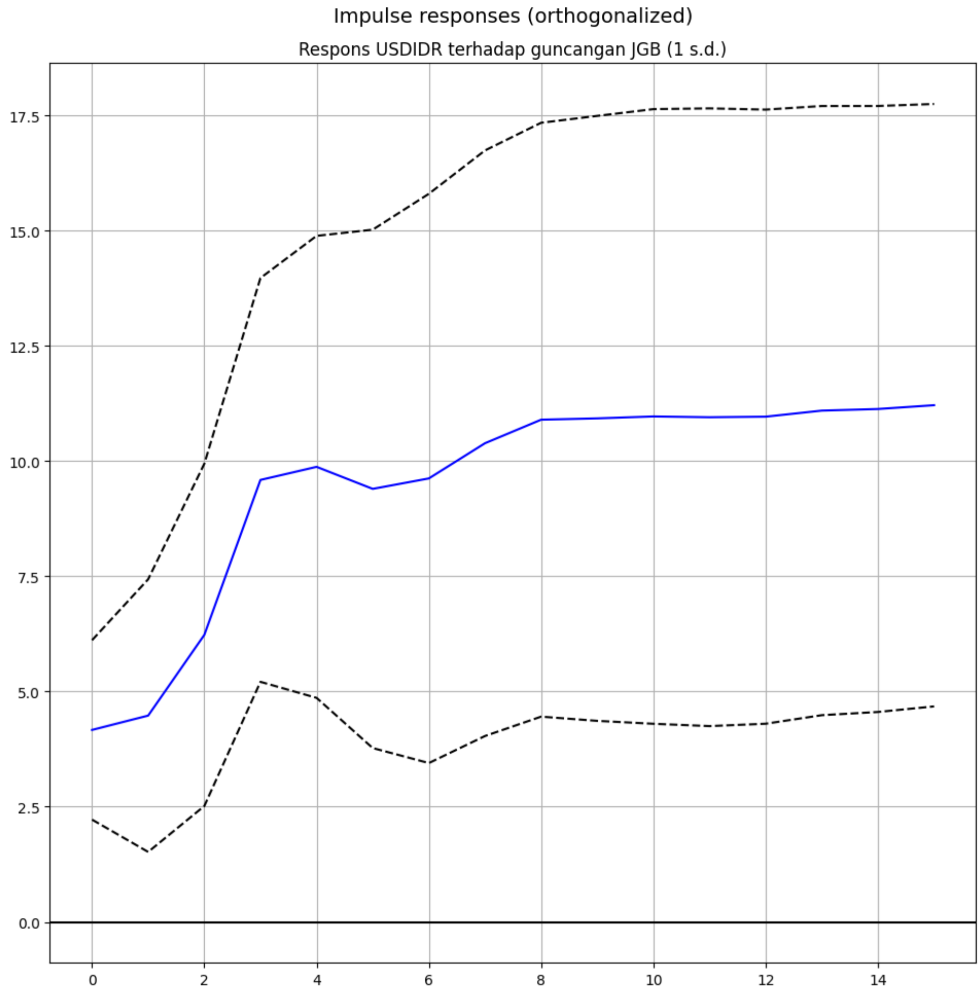

Figure 2: Orthogonalized response of USD/IDR to a one-standard-deviation JGB shock. The response is positive and persistent; the lower band remains above zero.

funding shock does not produce 18,000; a joint repricing does.

5 Robustness

The central findings hold up under the checks the results invite. Adding oil as a terms-of-trade control leaves the USD/IDR loading large and significant (14.4). Splitting the sample at the 2022 onset of BoJ normalization, the rupiah error-corrects strongly in both regimes, and the VIX's role *strengthens* in the post-2022 period (its loadings become significant in both relations), consistent with risk appetite mattering more as the carry unwind matured. The USD/IDR loading stays large and positive across lag orders of two to six and ranks of one and two.

Two checks speak to the near-$I(1)$ concern. In a fully stationary first-difference VAR, daily JGB changes Granger-cause daily SBN changes only weakly (p between 0.06 and 0.15). Far from undermining the result, this sharpens it: the transmission does not live

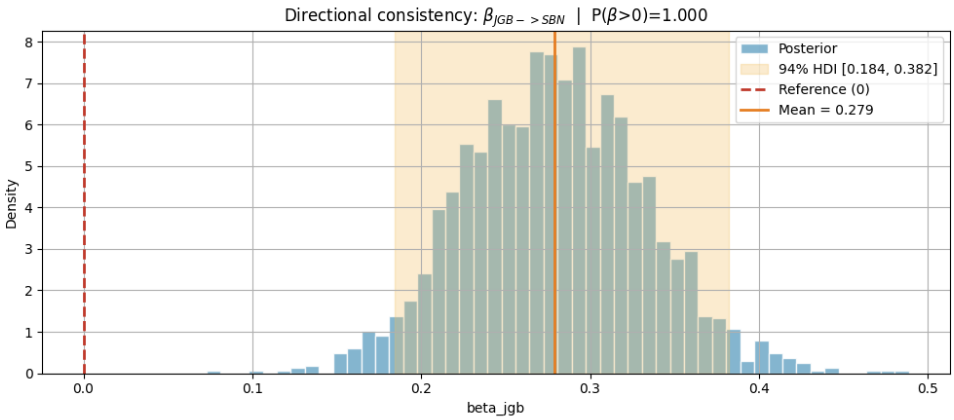

Figure 3: Posterior of the JGB→SBN daily slope. The mass lies entirely above zero (mean 0.28); the effect survives controlling for the dollar.

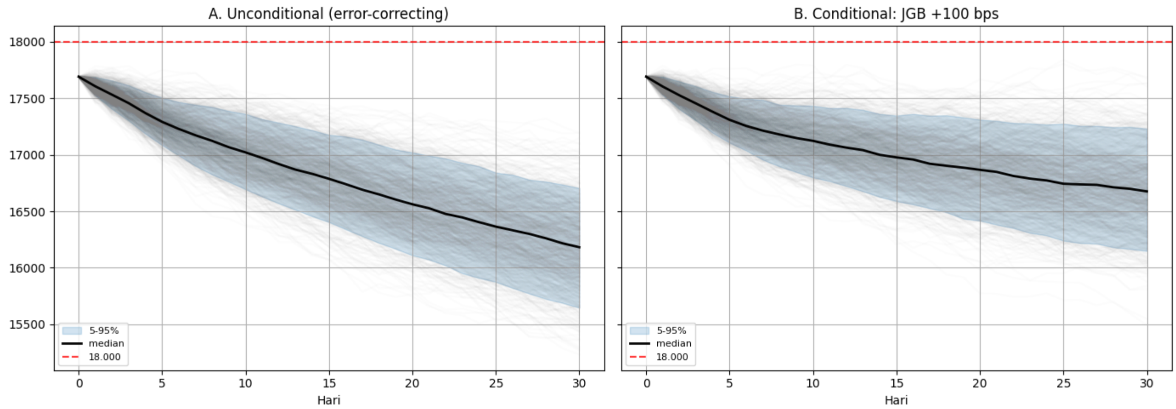

Figure 4: Monte Carlo from VECM dynamics. Left: unconditional paths error-correct toward equilibrium, well below the 18,000 line. Right: a conditional JGB-rise scenario shifts the median up but still falls short of 18,000.

in same-day, high-frequency co-movement but in the long-run equilibrium the VECM estimates. The contemporaneous Bayesian association and the weak predictive Granger link are not in tension—they say the relationship is a level relationship, not a daily one, which is exactly the case for cointegration over a returns regression.

We flag one limitation as future work. Reproducing the full magnitude of the 18,000 episode would require a joint stress scenario—JGB, dollar, and VIX moving together—or a post-2022-only equilibrium that may sit higher than the full-sample anchor. We leave that conditional-scenario exercise to an appendix in subsequent work; the present data already answer the question we set, which is the relative weight of the Japanese channel.

6 Conclusion

We set out to test whether rising Japanese yields drove the rupiah through the 2024–2026 carry unwind, and to do it with direct yield data rather than proxies. The transmission is real: the rupiah error-corrects, the impulse response is signed and persistent, and a Bayesian check puts almost all of the posterior mass on a positive JGB–SBN link. But the same analysis, properly weighted, shows the channel is small. A funding shock in Tokyo explains about 3% of rupiah variance even under the identification most favourable to it, against 30% for the dollar and a rising 15% for risk appetite—and the rupiah's equilibrium does not even exist as a stable object until risk appetite is written into it.

The honest summary is the one in our title. Japanese funding shocks contribute to rupiah dynamics, but their effect operates primarily through a broader equilibrium system dominated by global dollar conditions and risk appetite. The move toward 18,000 was not a JGB overshoot; it was a multi-factor repricing in which Japan was one of several forces and not the largest. For a policymaker the implication is practical: watching Tokyo is useful but insufficient, because the variable that organizes the rupiah's long-run equilibrium is the global price of risk. The limitations are the usual ones—the near-$I(1)$ status of two series, a variance decomposition that depends on ordering (which we addressed by handing the JGB the most generous position and finding it still small), and a single-country focus—but none of them disturb the central result, which is a matter of relative magnitudes rather than fragile significance.

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